

Exercise – Worm Kingdoms

“Slimy, yet satisfying”

Simba — The Lion King

After the lion cub’s arrival, everything changed for the grubs and worms of the jungle. Simba’s ferocious appetite had decimated their population, forcing them to put their differences aside. As a result, the grubs and worms decide to join their underground burrows to strengthen their defense. There are k burrows in total. Each burrow consists of a set of *rooms*. Each room r has a certain *size* s_r . Rooms are connected by *tunnels*, which can be used in both directions. Each existing tunnel connects two rooms within the same burrow. For two rooms r and r' , a *trip* from r to r' is a sequence $r_1, \dots, r_\ell$, for some $\ell \geq 1$, such that (1) $r_1 = r$; (2) $r_\ell = r'$; and (3) r_j and r_{j+1} are connected by a tunnel, for all $1 \leq j < \ell$. The burrows are designed such that for each pair of rooms (r, r') in the same burrow, there exists a unique trip from r to r' .

The reorganization requires a *nice partition*, that is, a partition of all rooms across all burrows into two groups W and G , to be managed by the worms and grubs, respectively, such that

- each group contains at least one room, and each room is assigned to exactly one group,
- for any two rooms r, r' within the same group, there exists a trip from r to r' , and
- for communication purposes, there is exactly one tunnel between W and G .

To facilitate the partitioning, new tunnels can be built, for a cost of p per tunnel. Ideally, the partition is also fair, in the sense that the amount of space assigned to both groups is about the same. To this end, the following cost function has been agreed upon:

$$\left| \sum_{r \in W} s_r - \sum_{r \in G} s_r \right| + p \cdot \# \text{new tunnels built}.$$

Can you help the grubs and worms determine the lowest cost achievable for a nice partition?

Input The first line of the input contains the number $t \leq 30$ of test cases. Each of the t test cases is described as follows:

- It starts with a line that contains two integers k p , separated by a space. They denote
 - k , the number of burrows ($1 \leq k \leq 300$) and
 - p , the cost of digging a new tunnel ($1 \leq p \leq 10^3$).
- Next follow k blocks of lines. The i -th block, for $0 \leq i < k$, defines the i -th burrow.
 - The first line of each block contains a single positive integer n_i , denoting the number of rooms in the burrow. You may assume $2 \leq \sum_{i=0}^{k-1} n_i \leq 300$.

- The next $n_i - 1$ lines describe the $n_i - 1$ tunnels of the burrow. Each line contains two integers u v , separated by a space, that indicate the existence of a tunnel between room u and v of the burrow ($0 \leq u, v < n_i$). You may assume that u lies on the unique trip from v to Room 0 of the burrow.
- The next n_i lines define the size of each room of the burrow. Each line contains a single integer. The number in the j -th line denotes the size s_j of the j -th room of the burrow ($1 \leq s_j \leq 2^8$).

Output For each test case output a single line with a single integer that denotes the lowest achievable cost of a nice partition.

Points There are four groups of test sets, worth 80 points in total. For each group there is also a corresponding hidden test set, worth 5 points. Overall, you can achieve 100 points.

1. For the first group of test sets, worth 25 points, you may assume that there is only one burrow ($k = 1$).
2. For the second group of test sets, worth 25 points, you may assume that every burrow comprises exactly one room ($n_i = 1$, for all $0 \leq i < k$).
3. For the third and fourth group of test sets, worth 20 and 10 points, respectively, there are no additional assumptions.

Sample Input	Sample Output
2	5
1 26	10
6	
0 1	
1 4	
1 5	
4 3	
5 2	
7	
4	
8	
5	
7	
10	
4 3	
1	
3	
1	
7	
1	
8	
1	
5	